Unit 2 Energy – Study Guide

You should know

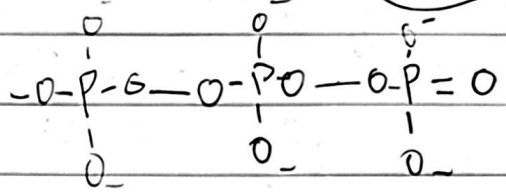
- How to interpret a graph of initial vs. final energy of reactants and products (i.e. is it showing a gain or loss of energy, and if it is occurring in one or multiple steps)
    - You should be able to *predict* whether a chemical reaction will require an input of energy or release energy, based on whether molecules are getting bigger or smaller
- Why ATP is an effective molecule for cells to use as energy currency
- Which class of molecules is the primary source of short-term energy for cells
- Which class of molecules serves as long term energy storage for cells
- The stages of cellular respiration, as well as what cells do when oxygen is unavailable
    - (Glycolysis, Krebs Cycle, Electron Transport Chain Fermentation)
    - You should be able to list the inputs and outputs of each stage (note, exact number of ATP is not important for ETC, but relative quantities)
        - You should be able to indicate what the products of fermentation are in our cells vs. bacterial and fungal cells
    - You should be able to describe *where* in the cell each process occurs
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- You should be able to label the following parts of mitochondria and chloroplasts (you should know these terms!)
    - Outer membrane, inner membrane, mitochondrial matrix, chloroplast stroma, thylakoid
- You should know the 5 domains of life, which stages of cellular respiration each uses, and which ones utilize photosynthesis
    - You should be able to explain why the endosymbiotic theory claims mitochondria and chloroplasts were once free living bacterial cells

Cellular Respiration

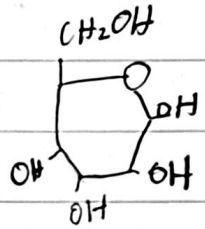
input
- Food
- Photosynthesis
making the molecules

storage
- short term: carbs
- long term: fats

active use



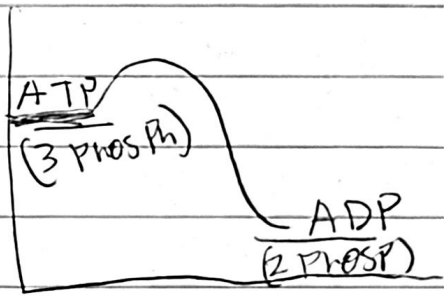
Glucose



6 carbon sugar

+2 ATP

2x Pyruvate (3 carbon sugar)



exothermic process

(cellular respiration: process of turning carbs to ATP.

3 parts of cellular respiration:

- Glycolysis → Fermentation (does not get more energy)
- Krebs Cycle ← +2 ATP
- e⁻ transport chain

$C_6H_{12}O_6$

$(C_3H_3O_3^-) \times 2$

2 Pyruvate (3 carbon sugar)

+2 ATP

$2NAD^+ \rightarrow 2NADH$

GLY: input = glucose

(2 NAD^+
2 ADP + 2 Phosphates)

→ 2 pyruvate (3-C sugar)
- 2 ATP (+2 H_2O)
- 2 NADH

① O_2 + mitochondria
pyruvate → Krebs cycle

② Fermentation
pyruvate → ethanol (Fungi: yeast, bacteria)
Lactic cells (animal cells)

(molecule)

- ATP is the energy carrier OF cells

Glucose —2 ATP→ 38 ATP

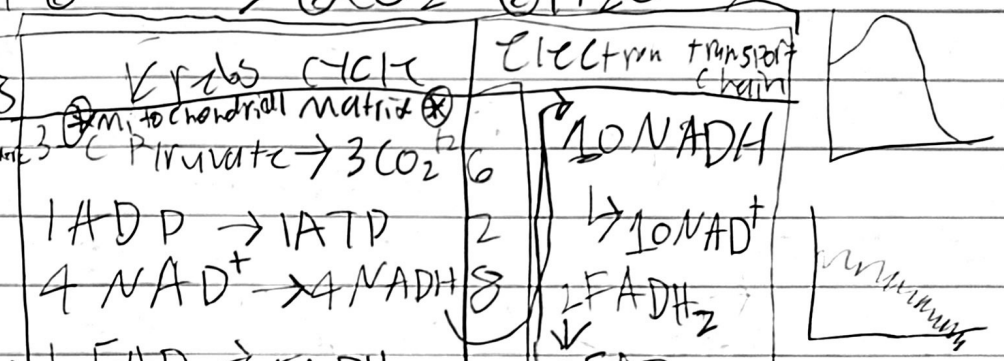
$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ (IF O_2 is present:)

GLYCOLYSIS
- Cytoplasm (entire in eukaryotic interior)
- 6 Carbon Glucose →
2 X 3-Carbon Pyruvates
2 ADP → 2 ATP
2 NAD^+ → 2 NADH
2 NAD—H
2 electrons
releases H^+ and 2 e^- separately

Krebs CYCLE
- Mitochondrial matrix
2 Pyruvate → $3CO_2$ (6)
1 ADP → 1 ATP (2)
4 NAD^+ → 4 NADH (8)
1 FAD → $FADH_2$ (2)

Electron transport chain
10 NADH → 10 NAD^+
2 $FADH_2$ → 2 FAD
34 ADP → 34 ATP
$O_2 \rightarrow H_2O$

most ATP comes at the end (etc)



no O_2:
Fermentation
regenerate NAD^+
no further energy



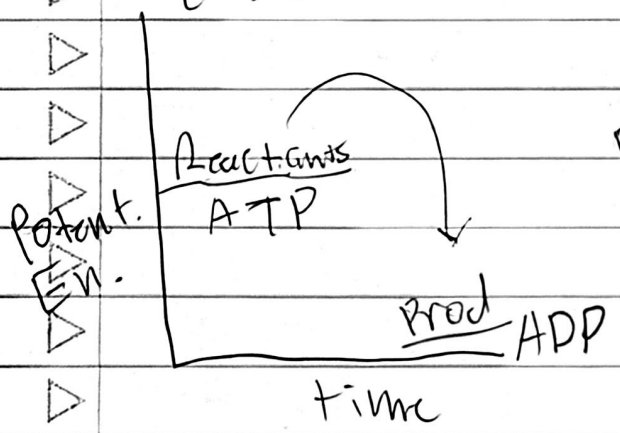
ATP Synthase: protein structure that looks like a motor. makes ATP when it spins

Food → use sunlight to build them → lipids (Fats) carbohydrates → ATP → Cellular respiration

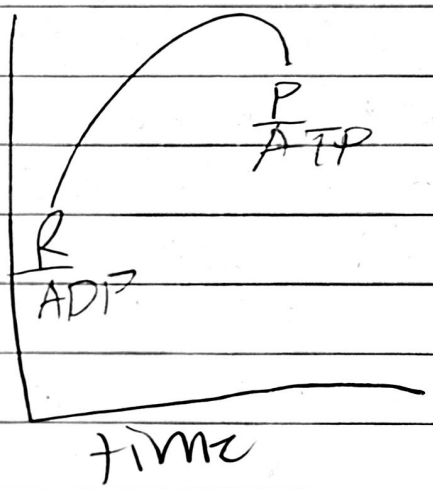
obtain | store | use | Energy

obtain	store	use
Sales	- bonds	Cash
labor	- stocks	- available for transactions
inheritance	R.E. Value	
Can be in transactable Form or in Stored form.	• stored value needs to be converted in Form to be used.	

Exothermic
(−) Change in Pot. E



Endothermic
(+) Change in Pot. E



Change Final - initial

ATP = Adenosine TriPhosphate → Cellular Cash



high - PotE
b/c (−) Charges close together.

Emet Classical Academy

10th Grade Integrated Science II

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Cellular Respiration

mitochondrion matrix

1. Glycolysis
2. Krebs Cycle
3. Elec. Trans. Chain (ETC)

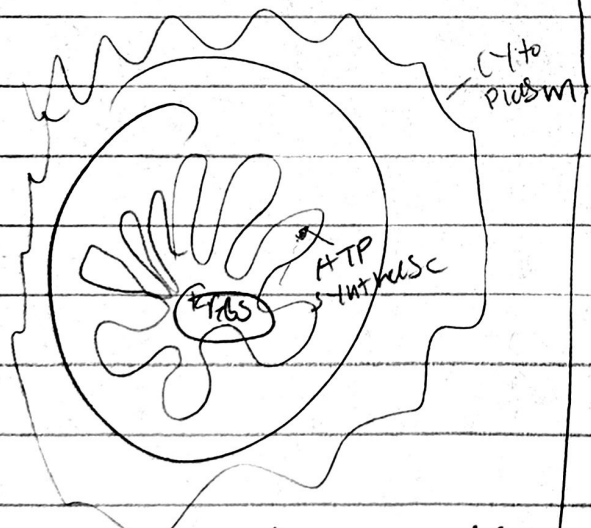
Big ↓ Small

(1) $C_6H_{12}O_6$ (glucose) ↓ Pyruvate ↓ (2) CO_2

(ATP) e^- → e^- NADH, $FADH_2$ carry e^- to ETC

e^- + ATP! ↓ Oxygen is the acceptor

(3) O_2 ↓ H_2O



inner mitochondrial membrane

Photosynthesis

(3) Calvin Cycle - Stroma of chloroplasts

(2) ETC - along the thylakoid membrane

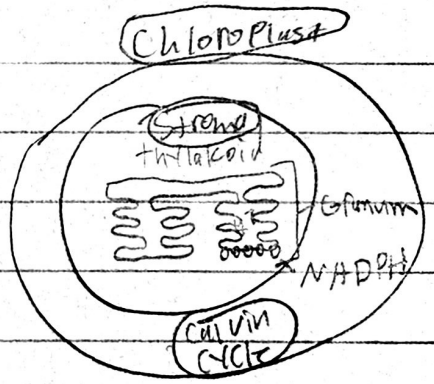
(1) light-dependent reaction - thylakoid membrane gives them to glucose

Small ↓ Big to carry electrons

(3) $C_6H_{12}O_6$ ↑ NADPH is electron acceptor stores ATP!

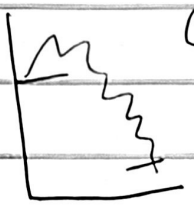
(2) CO_2 ↑ O_2 ↑ H_2O

e^- (3)





all energy transfers lose energy

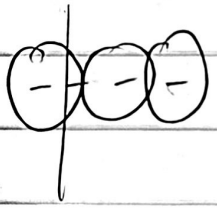


capture it as ATP

Sun = external input of energy

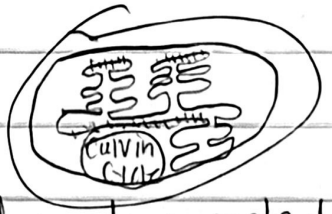
Big → small molec molec → release energy (spontaneous)

Photosynthesis: using light to make big molecules



unstable:

Chloroplast * only in photosynthetic orgs
- plants
- some bacteria (cyanobacteria)



light dependent - double membrane
- thylakoid

Chlorophyll
- a protein that captures light energy
- use it to split $2H_2O$ into O_2 and $4H^+$

$2NADP^+ \rightarrow 2NADPH$ $16e^-$

$12e^-$ $4e^-$

2 NADPH → electron transport

(electron carrier) | pump H^+ to create a gradient

Calvin cycle | H^+ flows back through ATP Synthase and makes ATP

$CO_2 + H_2O \rightarrow C_6H_{12}O_6$

3, 9 molecule

phosphate group alone attached to a diacylglycerol does not qualify as a phospholipid. It is phosphatidate (diacylglycerol 3-phosphate), the precursor of phospholipids. An alcohol modifies the phosphate group. Phosphatidylcholine and phosphatidylserine are two important phospholipids that are in plasma membranes.

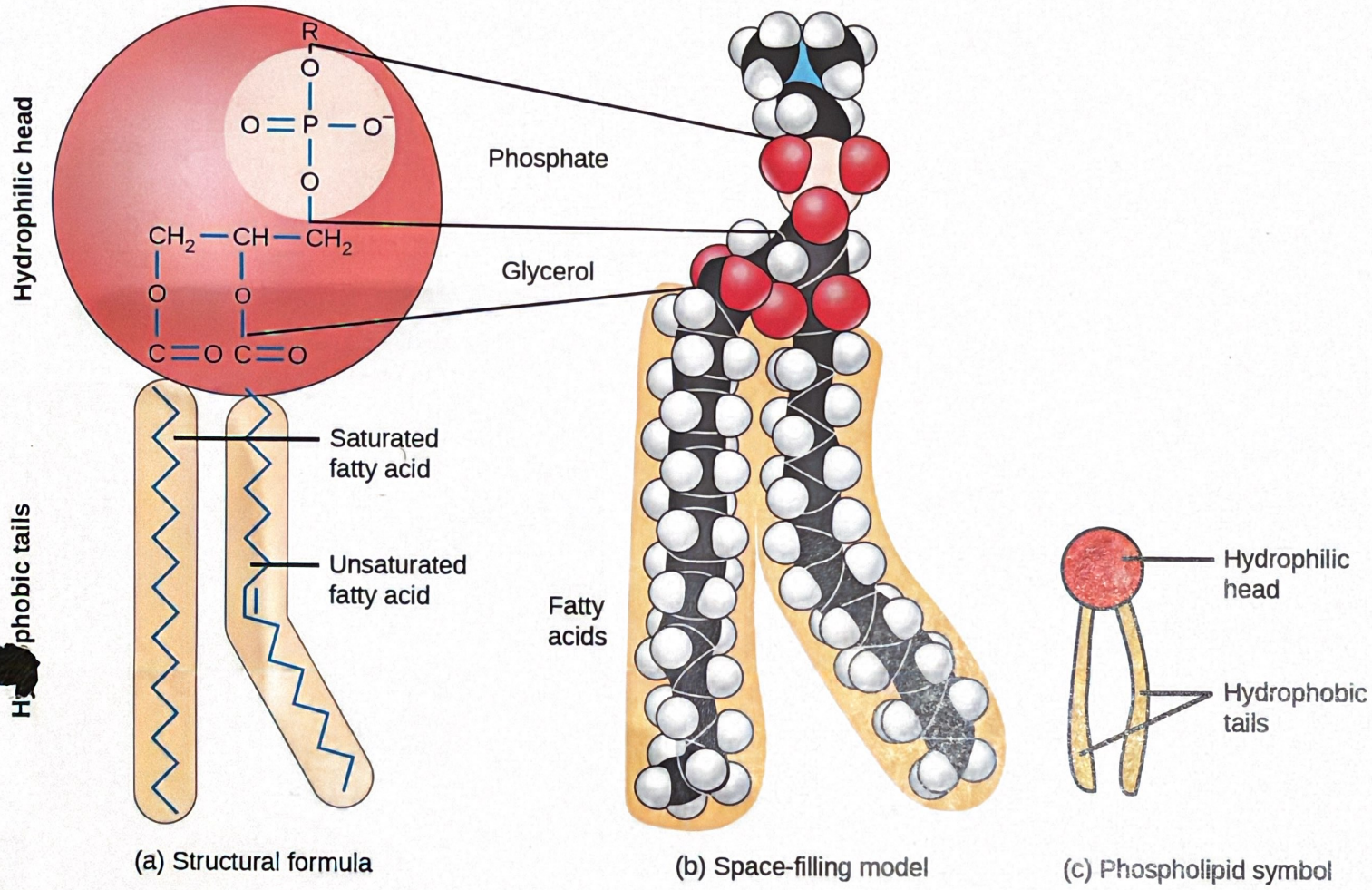
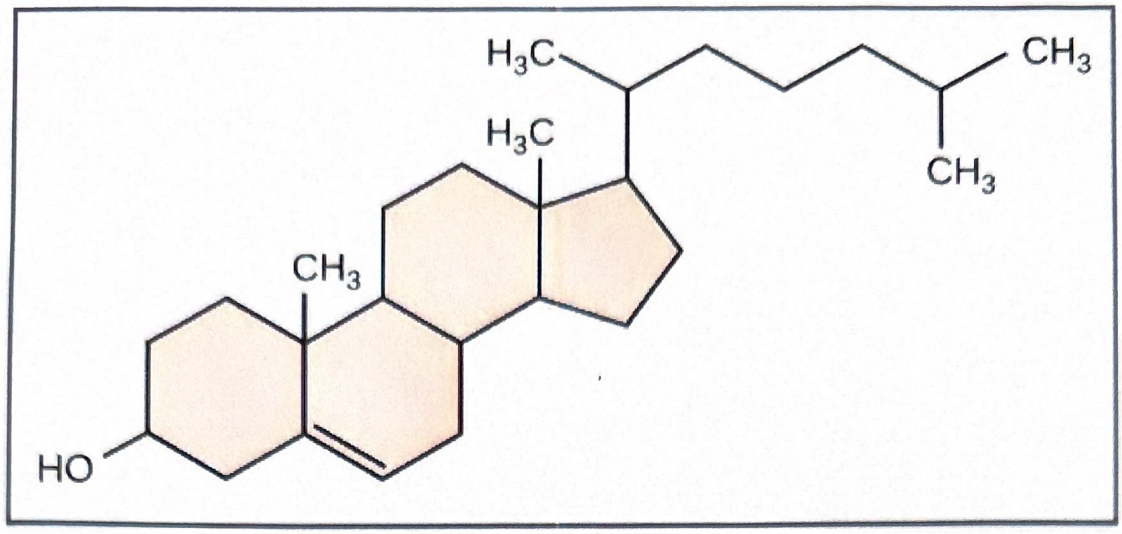
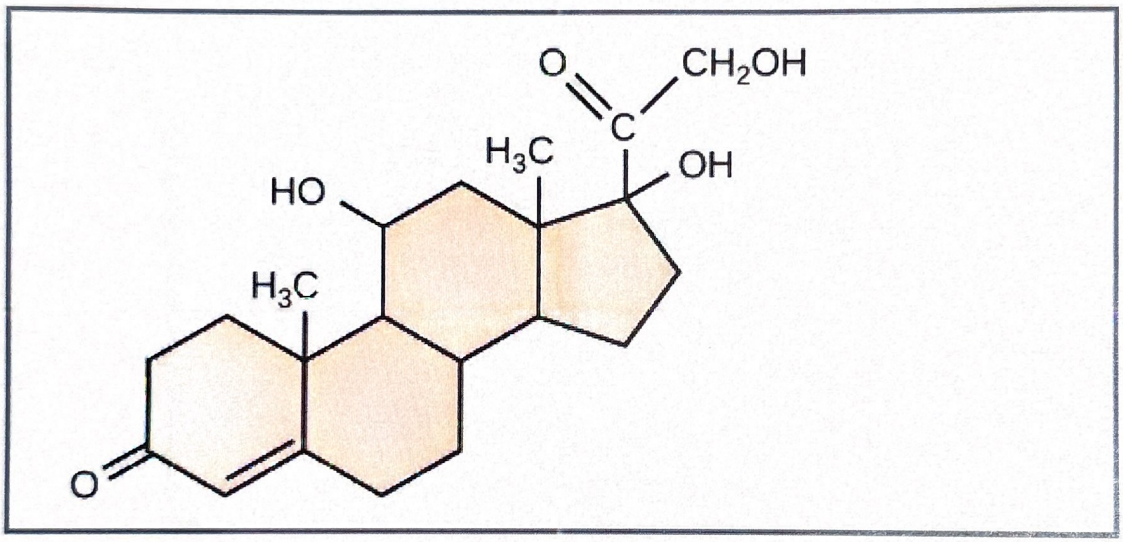


Figure 3.19 A phospholipid is a molecule with two fatty acids and a modified phosphate group attached to a glycerol backbone. Adding a charged or polar chemical group may modify the phosphate.

A phospholipid is an amphipathic molecule, meaning it has a hydrophobic and a hydrophilic part. The fatty acid chains are hydrophobic and cannot interact with water; whereas, the phosphate-containing group is hydrophilic and interacts with water (Figure 3.20).



Cholesterol



Cortisol

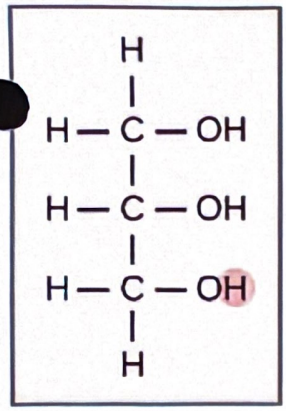
Figure 3.21 Four fused hydrocarbon rings comprise steroids such as cholesterol and cortisol.

Cholesterol is the most common steroid. The liver synthesizes cholesterol and is the precursor to many steroid hormones such as testosterone and estradiol, which gonads and endocrine glands secrete. It is also the precursor to Vitamin D. Cholesterol is also the precursor of bile salts, which help emulsifying fats and their subsequent absorption by cells. Although lay people often speak negatively about cholesterol, it is necessary for the body's proper functioning. Sterols (cholesterol in animal cells, phytosterol in plants) are components of the plasma membrane of cells and are found within the phospholipid bilayer.

LINK TO LEARNING

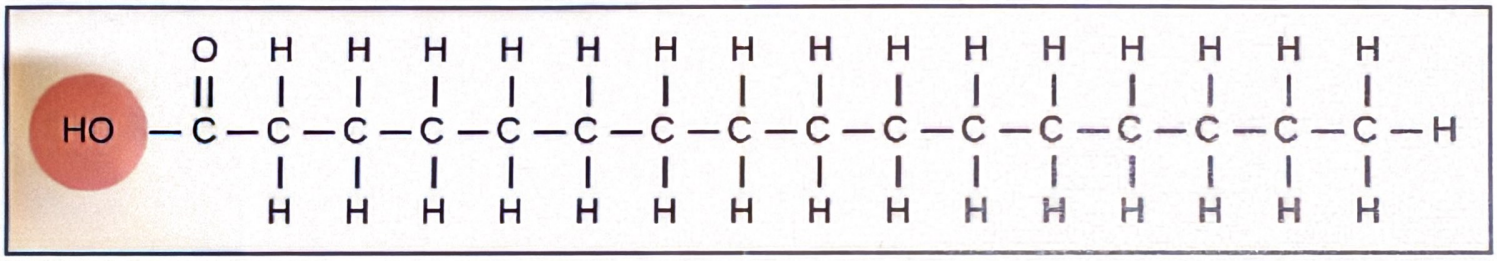
For an additional perspective on lipids, watch this video about types of fat.

Glycerol

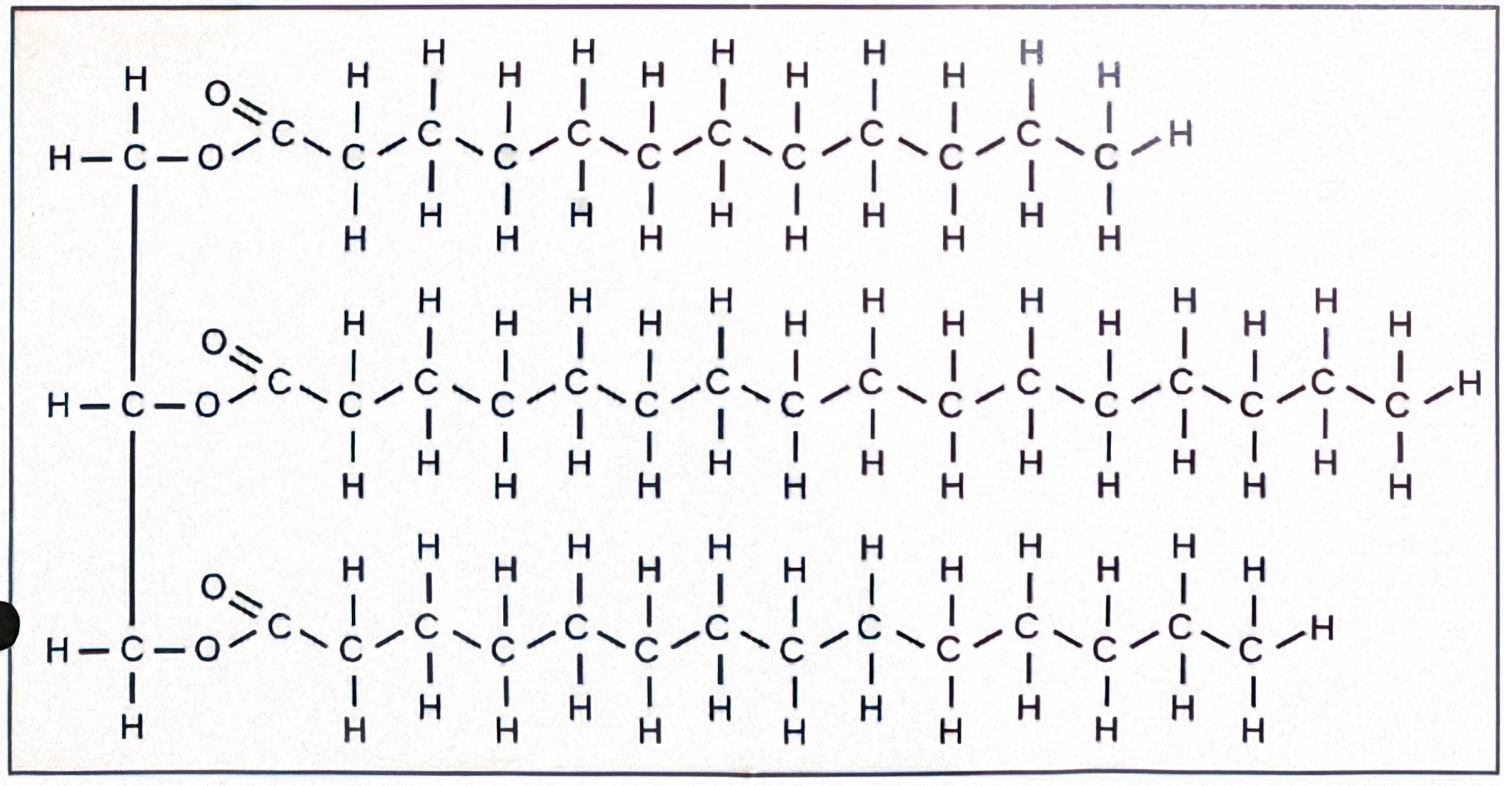


+

Fatty Acid



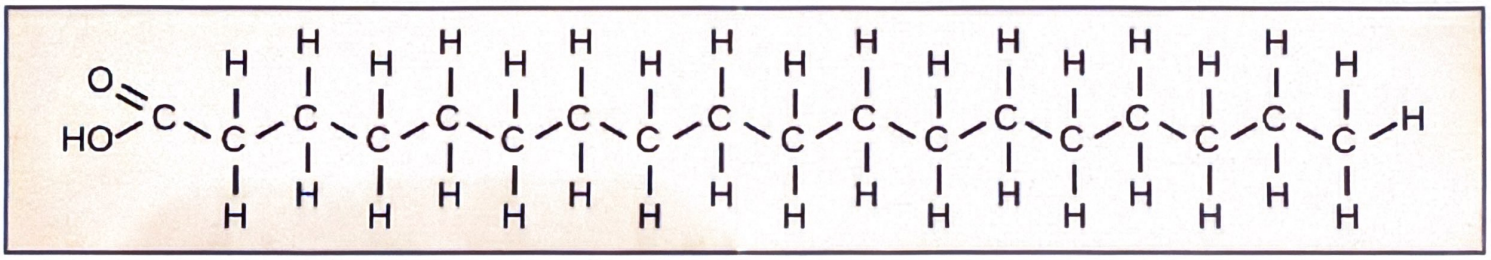
Triacylglycerol



molecule around the double bond. If hydrogens are present in the same plane, it is a cis fat. If the hydrogen atoms are on two different planes, it is a **trans fat**. The *cis* double bond causes a bend or a "kink" that prevents the fatty acids from packing tightly, keeping them liquid at room temperature (Figure 3.16). Olive oil, corn oil, canola oil, and cod liver oil are examples of unsaturated fats. Unsaturated fats help to lower blood cholesterol levels; whereas, saturated fats might contribute to plaque formation in the arteries.

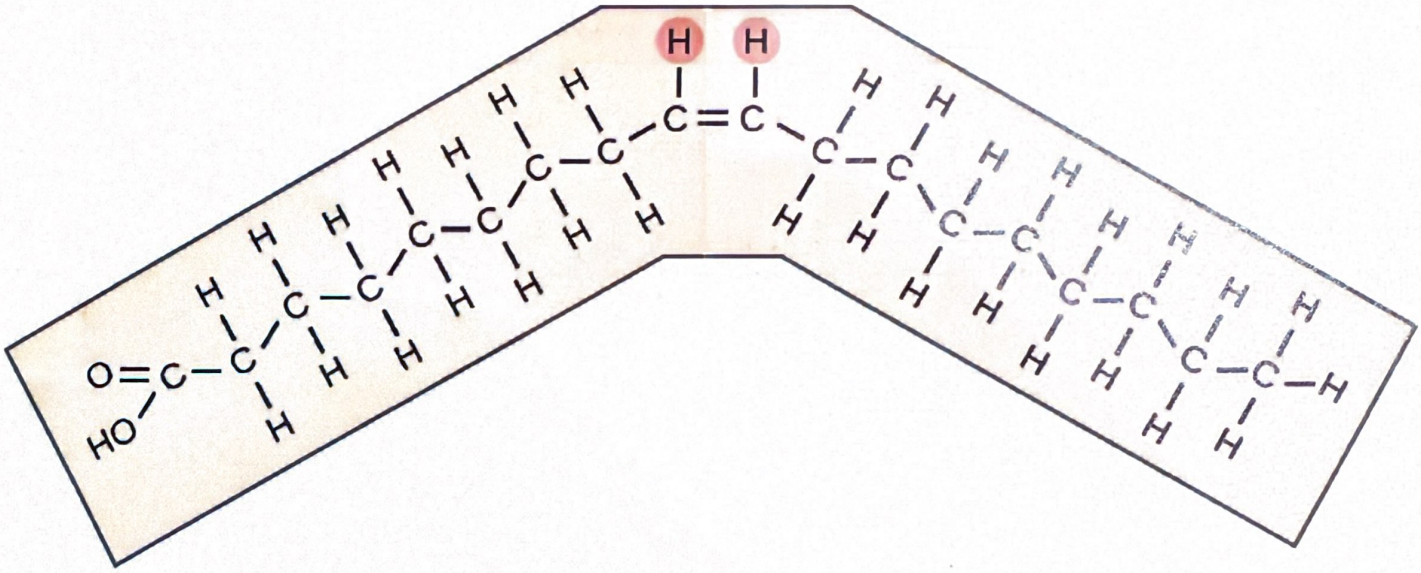
Saturated fatty acid

Stearic acid



Unsaturated fatty acids

Cis oleic acid



Trans oleic acid

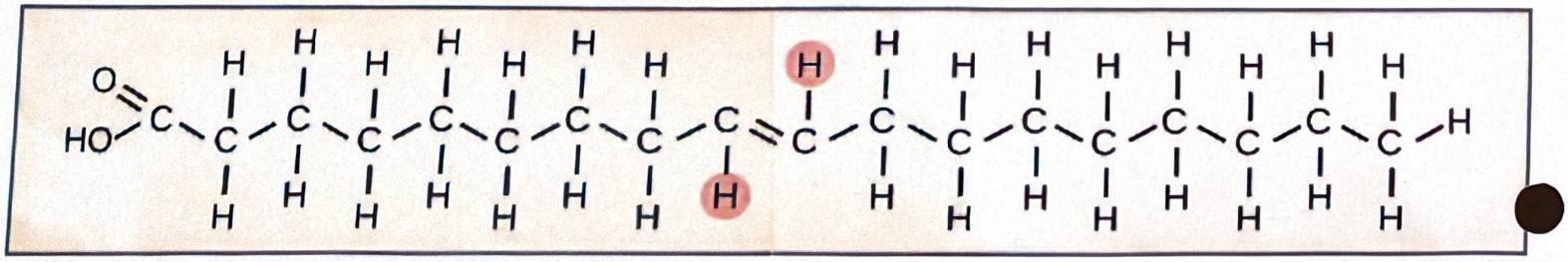
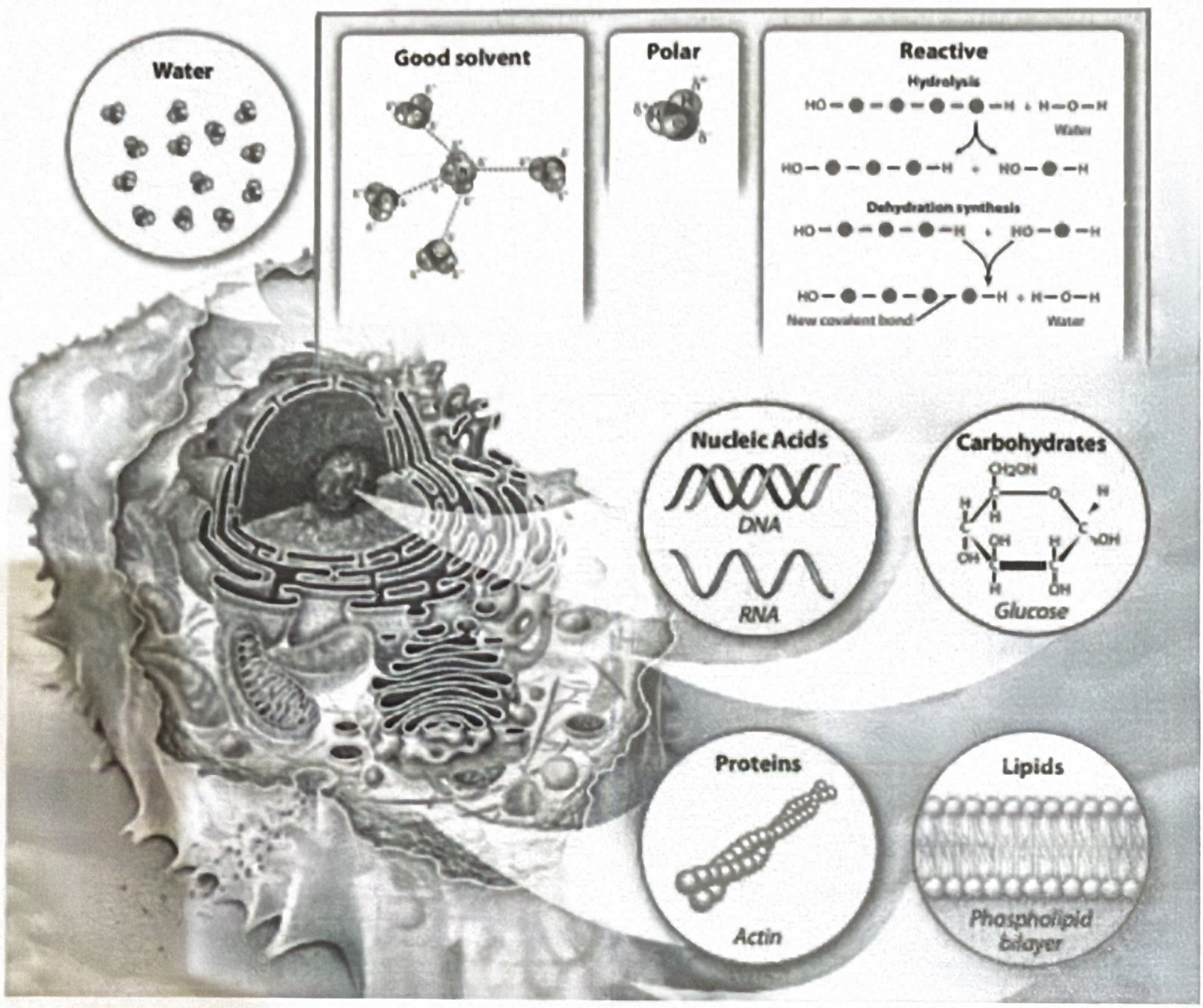


Figure 3.16 Saturated fatty acids have hydrocarbon chains connected by single bonds only. Unsaturated fatty acids have one or more double bonds. Each double bond may be in a *cis* or *trans* configuration. In the *cis* configuration, both hydrogens are on the same side of

Water

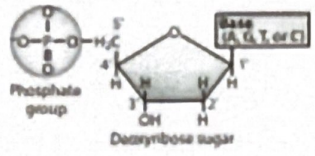

Water
Good solvent
Polar
Reactive
Hydrolysis
Water
Dehydration synthesis
New covalent bond
Water
Nucleic Acids
DNA
RNA
Carbohydrates
Glucose
Proteins
Actin
Lipids
Phospholipid bilayer

Organic Molecules

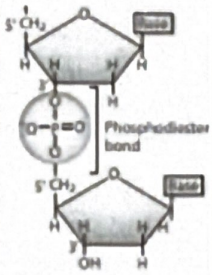
Nucleic Acids

Functions: Information storage and transmission, catalysis (RNA)
Examples: DNA, RNA

Subunits: Nucleotides



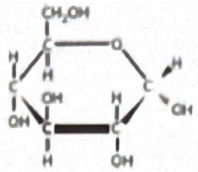
Bonds: Phosphodiester



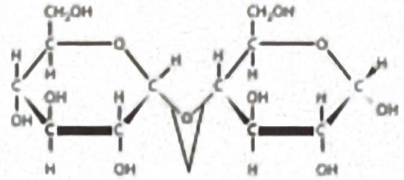
Carbohydrates

Functions: Energy storage, cell identification
Examples: Sugars, cell wall, exoskeleton

Subunits: Monosaccharides



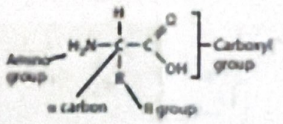
Bonds: Glycosidic



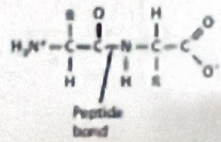
Proteins

Functions: Cell structure, signaling, and catalysis
Examples: Cytoskeleton, signaling molecules, enzymes

Subunits: Amino Acids



Bonds: Peptide

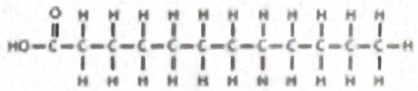


Lipids

Non-polar

Functions: Plasma membrane, energy storage, cell signaling
Examples: Phospholipids, triacylglycerol, cholesterol, steroids

Bonds: Many carbon-carbon and carbon-hydrogen bonds



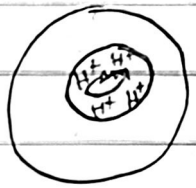
Eukaryotes

DNA enclosed in a nucleus

membrane bound organelles
- mitochondria
- chloroplast

Prokaryotes

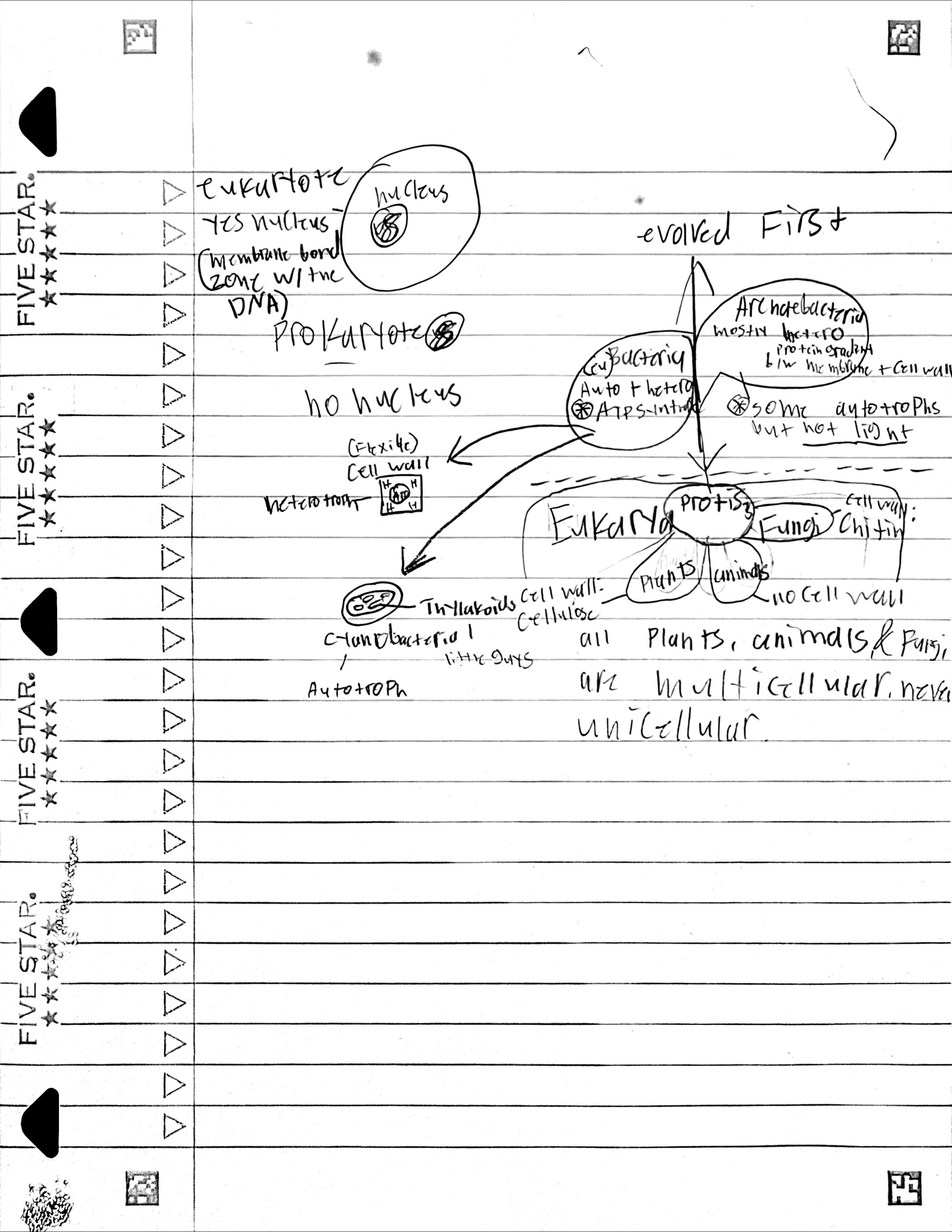
Circular DNA Floating



Heterotrophs
- eat their food (hunt)

Autotrophs
- make their own food

Endosymbiosis - Eukaryotes arose by engulfing another bacteria which became the mitochondria


FIVE STAR
FIVE STAR
FIVE STAR
FIVE STAR
Eukaryote
Yes nucleus
(membrane bord zone w/ the DNA)
nucleus
Prokaryote
no nucleus
(Extra) Cell wall
heterotroph
Cyanobacteria
little guys
Autotroph
Thylakoids
evolved First
Archaebacteria
mostly hetero
protein coating
b/w the membrane + Cell wall
Eubacteria
Auto + hetero
ATP-synth
some autotrophs
but not light
Eukarya
protists
Fungi
Cell wall: Chitin
Plants
animals
no Cell wall
Cell wall: Cellulose
all Plants, animals & Fungi are multicellular, never unicellular.